

SHAHADAT HUSSAIN, PhD

MATERIALS & MANUFACTURING ENGINEER — ADDITIVE MANUFACTURING / METAMATERIALS

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PROFESSIONAL SUMMARY

Materials and manufacturing engineer with 6+ years of experience in additive manufacturing (LPBF, FFF, SLA) process development, materials qualification, and mechanical/thermal testing across advanced research and applied engineering settings in the UAE. Hands-on operator of 10+ industrial 3D printing systems and a full metallurgical characterization and testing suite (SEM, XRD, DSC, Instron, MTS). Specialist in architected lattice structures (TPMS) and shape memory alloys, with direct relevance to lightweight, thermally efficient, and vibration/fatigue-resistant component design. Combines materials engineering depth with Python/MATLAB-based data analysis to drive process optimization, defect reduction, and quality mapping. Seeking to bring this qualification, testing, and process-engineering expertise to manufacturing engineering or materials qualification roles supporting power and thermal management hardware.

CORE COMPETENCIES

- Additive Manufacturing Process Development (LPBF, FFF, SLA)
- Mechanical & Thermal Qualification Testing (Static, Fatigue, Compression, Vibration/Modal)
- Materials Characterization (SEM, XRD, DSC, Optical Microscopy)
- Architected Lattice Structures for Thermal & Structural Performance (TPMS)
- Process Parameter Optimization & Defect/Quality Mapping
- NiTi & Cu-Based Shape Memory Alloys
- CAD & Build Preparation (SolidWorks, FreeCAD, Materialise Magics, GrabCAD)
- Data Analysis & Process Modeling (Python, MATLAB, SQL)
- EHS / Lab Safety & Risk Assessment

PROFESSIONAL EXPERIENCE

Research Associate — Additive Manufacturing & Advanced Materials

Sep 2024 – Present

New York University Abu Dhabi (NYUAD), Abu Dhabi, UAE

- Delivered carbon-fiber-reinforced PLA/PPA composite lattice components (gyroid and primitive TPMS geometries) via FFF, engineered for lightweight, high-strength, vibration-damping performance relevant to structural and thermal hardware applications.
- Qualified printed composite structures against quasi-static compression, flexural, modal, and frequency-response test protocols to validate strength, energy absorption, and vibration performance under service-like loading.
- Operated and maintained 10+ industrial additive manufacturing systems (Raise3D Pro3 Plus HS, Stratasys J850/F370, Formlabs Form3, Bambu Lab Carbon X1), sustaining continuous production and prototyping throughput.

Postdoctoral Research Fellow — Metal 3D Printing & Smart Materials

Feb 2020 – Jun 2024

Khalifa University, Abu Dhabi, UAE

- Fabricated and qualified NiTi shape memory alloy TPMS lattice structures via Laser Powder Bed Fusion (EOS M400); optimized laser power, scan speed, and energy density to control microstructure, porosity, and phase transformation behavior — a process-control approach directly transferable to defect reduction in production manufacturing.
- Executed a full materials characterization pipeline on 50+ printed samples using SEM, FIB-SEM (SEI/BSE/EDS), XRD, DSC, and optical microscopy to link process parameters to structural and functional performance.
- Ran static (Instron 5969, 50 kN), fatigue (Instron 8872, 25 kN), and compression (MTS 100 kN) qualification campaigns; developed defect and print-quality maps used to reduce porosity and improve part consistency — a qualification methodology applicable to reliability testing of power and thermal hardware.
- Authored 4 peer-reviewed publications; completed comprehensive EHS lab safety training covering chemical handling, cryogenic substances, risk assessment, and emergency response protocols.

Project Fellow — Smart & Functional Materials

Jun 2013 – Aug 2014

CSIR – Advanced Materials & Processes Research Institute (AMPRI), Bhopal, India

- Synthesized Cu-Al-Mn and Cu-Al-Ni shape memory alloys via vacuum induction melting under a government-funded national research program; performed hot/cold rolling, heat treatment, quenching, and metallographic preparation.
- Characterized phase transformation and mechanical properties using SEM, XRD, DSC, and hardness/tensile testing; contributed to 5 peer-reviewed publications and a research collaboration with IIT Madras.

Engineering Intern — Steel Plant Operations

Jun – Jul 2011

Bokaro Steel Plant, Steel Authority of India Limited (SAIL), Jharkhand, India

- Gained direct industrial exposure to large-scale integrated steel manufacturing processes, plant operations, and quality systems.

EDUCATION

PhD, Materials Science & Technology | 2014 – 2019

Academy of Scientific and Innovative Research (AcSIR) / CSIR-AMPRI, Bhopal, India — Thesis: Cu-based shape memory alloys: composition, processing, microstructure, and shape memory properties

BEng, Mechanical Engineering | 2008 – 2012

UIT, Rajiv Gandhi Proudyogiki Vishwavidyalaya (RGPV), Bhopal, India

TECHNICAL SKILLS

Additive Manufacturing Systems: EOS M400 (LPBF/SLM); Stratasys F370/J850 (FDM/PolyJet); Raise3D Pro3 Plus HS, Forge1 (FFF); Formlabs Form3 (SLA); Ultimaker S1

Materials Characterization: JEOL 7610F SEM; FEI Nova NanoSEM; FEI Quanta 3D FIB-SEM (SEI/BSE/EDS); Bruker D2 & Rigaku MiniFlex XRD; Mettler Toledo & Setaram DSC; Leica Optical Microscopy

Mechanical Testing: Instron 5969 (static, 50 kN); Instron 8872 (fatigue, 25 kN); MTS 100 kN UTM; Controls Simplex 350 & Automax

Design & Computing: SolidWorks, FreeCAD, Materialise Magics, GrabCAD Print; Python (Pandas, NumPy, Scikit-learn), MATLAB, SQL, Linux

CERTIFICATIONS

- Additive Manufacturing Specialization — Arizona State University
- IBM Data Science Professional Certificate — IBM
- Google AI Essentials Certificate — Google
- Lab Safety (EHS) — Khalifa University

PUBLICATIONS & RECOGNITION

15+ peer-reviewed publications | 500+ citations | h-index: 10 | Peer reviewer for 15+ journals including Additive Manufacturing, Materials & Design, Scripta Materialia. Full list: scholar.google.com/citations?user=tQNSWaAAAAAJ

ADDITIONAL INFORMATION

Languages: English (Professional), Hindi (Native), Urdu (Native) | Profiles: github.com/drshahadathussain | researchgate.net/profile/Shahadat-Hussain

SHAHADAT HUSSAIN, PhD

MATERIALS CHARACTERIZATION | PHYSICAL METALLURGY | MECHANICAL TESTING

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August 26, 2026

Hiring Manager

Evolito

Bicester, UK

Subject: Senior Materials Engineer – Job Requisition JR100190

Dear Hiring Manager,

I am writing to express my enthusiastic interest in the Senior Materials Engineer position at Evolito. Having followed Evolito's mission to accelerate the adoption of electric propulsion for a net-zero world, I am excited by the prospect of contributing my deep expertise in metallic materials, advanced characterization, and additive manufacturing to support your airworthiness certification programmes and ambitious growth plans.

In my current role as a Research Associate at NYU Abu Dhabi, and throughout my Ph.D. in Materials Science, I have dedicated my career to unraveling complex "process-structure-property" relationships in high-performance materials. My core expertise lies in the characterization of advanced metallic alloys (including NiTi and Cu-Al-Ni shape memory alloys) using a comprehensive suite of analytical techniques—including SEM, EBSD, XRD, and thermal analysis (DSC). I routinely design and execute experimental programmes to determine thermal and mechanical properties, making me a natural fit for your need to manage materials characterisation activities. Furthermore, my extensive hands-on experience with Laser Powder Bed Fusion (LPBF) and Fused Filament Fabrication (FFF) aligns directly with the manufacturing processes vital to producing lightweight, high-torque-density electric propulsion systems.

While my specialized research focuses on metals, polymers and alloys, my fundamental training in materials science provides a robust understanding of broader material classes, including composites, and adhesives. I am confident in my ability to rapidly apply my systematic, analytical mindset to evaluate these materials across their full product lifecycle, develop robust test procedures, and support your design and manufacturing teams during new process transfers. My academic and industry collaborations have honed my failure investigation skills, giving me a rigorous, data-driven approach to root cause analysis that ensures corrective actions are both technically sound and practically implementable.

Evolito's requirement for a self-starting, collaborative professional resonates strongly with my work ethic. I thrive in dynamic, fast-moving environments and have a proven ability to work cross-functionally with suppliers and internal stakeholders to translate complex research into tangible engineering solutions. Regarding logistics, I am currently based in the UAE but am fully prepared and eager to relocate permanently to Bicester. I am committed to securing my full UK driving licence upon arrival to ensure I can travel as required for the role.

I am deeply inspired by the prospect of joining Evolito during this high-growth phase and applying my skills in materials characterisation and processing to help revolutionize personal mobility and clean aviation. I would welcome the opportunity to discuss how my background can contribute to your team's success.

Thank you for your time and consideration.

Sincerely,

Shahadat Hussain, PhD